

Database Management Systems (DBMS) - Sample Notes

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Introduction

A Database Management System (DBMS) is software that allows users to create, manage, and retrieve data efficiently. Examples include PostgreSQL, MySQL, Oracle Database, and SQLite.

Normalization

Normalization is the process of organizing data to reduce redundancy and improve data integrity. The common normal forms are 1NF, 2NF, 3NF, and BCNF.

SQL Commands

DDL commands include CREATE, ALTER, and DROP. DML commands include INSERT, UPDATE, and DELETE. DQL uses the SELECT statement for querying data.

Transactions

A transaction is a sequence of operations performed as a single logical unit of work. Transactions follow the ACID properties: Atomicity, Consistency, Isolation, and Durability.

Indexing

Indexes improve query performance by allowing faster access to rows. Common index structures include B-Tree and Hash indexes.

Distributed Databases

Distributed databases store data across multiple locations while appearing as a single database system.